

Component 4: Load Month

xPostForecast · Milestone 4: Component-Level Design · Pseudocode

Where it runs Backend: loadMonth, readGrid, samplePoints (backend/routes/stac.js, backend/sign/sign.js)

User stories US-6, US-7, US-8

Flowchart Component4_LoadMonth_Flowchart.pdf

Description

Gets one month of NOAA nClimGrid average temperatures from Microsoft Planetary Computer and turns the grid cells inside West Virginia into map points. Returns the points, NONE if the month isn't in the dataset, or fails if the outside service has a problem.

Design principles

Abstraction and stepwise refinement. To its caller, LoadMonth is one abstract step: "get this month's points." Inside, that step is refined into smaller ones (search the catalog, sign the file address, download, read the West Virginia window, convert each cell), and the signing step is refined again into its own module, UrlSigner.

Information hiding. Everything specific to Planetary Computer (catalog address, collection name, the "tavg" asset, access tokens, the GeoTIFF file format) is hidden inside this component. If the data source changed, only this component would change; Component 3 and the frontend would not.

Pseudocode

```
BEGIN LoadMonth(month)                    returns points, NONE, or FAILS
  SEARCH the Planetary Computer catalog for collection noaa-ncimgrid-monthly,
    inside West Virginia's bounding box, at the 15th of month

  IF the search request failed THEN
    FAIL "catalog search failed"
  END IF

  IF the search found no item THEN
    RETURN NONE
  END IF

  SET asset TO the item's "tavg" (average temperature) asset

  IF the item has no "tavg" asset THEN
    FAIL "item has no tavg asset"
  END IF

  CALL Sign(asset address), GIVING signedAddress

  IF signing failed THEN
    FAIL "signing failed"
  END IF

  DOWNLOAD the file at signedAddress
```

```

IF the download failed THEN
    FAIL "download failed"
END IF

READ the cell values (°C) for the rows and columns that cover the bounding box
SET points TO an empty list

FOR EACH cell in the window, row by row
    IF the cell's value is a no-data value THEN
        SKIP to the next cell
    END IF

    SET lon, lat TO the center of the cell

    IF (lon, lat) is outside the bounding box THEN
        SKIP to the next cell
    END IF

    ADD a point (lat, lon, tavg = value × 9 / 5 + 32) TO points

    IF points holds 10,000 points THEN
        RETURN points
    END IF
END FOR

RETURN points
END LoadMonth

```